QPCODE: 25900043

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (Honours) REGULAR EXAMINATION, MARCH 2025

SECOND SEMESTER

Core Course (CC) - MG2CCRBCA102 - OPERATING SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skills(S), Interest(I), and Appreciation(Ap)

SCHEME FOR VALUATION

Part A

Very Short Answer Questions. Answer all questions
Each question carries 2 marks

1. **Identify one key function of the Timer in an operating system [K] / [CO1]**
   The OS maintains control over the system and prevents user programs from running indefinitely in user mode or Kernel mode.
2. **What is the purpose of the Process Control Block (PCB)? [U] / [CO2]**
   Each process is represented in the operating system by a process control block (PCB). It contains many information associated with a specific process, Including process state, program counter, CPU scheduling information, CPU registers etc.
3. **Explain the necessary conditions required for deadlock to occur. [U] / [CO3]**
   The four necessary conditions for the occurrence of deadlock are as follows:
   1) Mutual Exclusion 2) Hold and Wait 3) No Preemption 4) Circular Wait
4. **How can a process execute its critical section? [U] / [CO3]**
   When one process is executing in its critical section, no other process is allowed to execute in its critical section. Each process must request permission to enter its critical section. The section of code implementing this request is the entry section, exit section and the remainder section.
5. **What is internal fragmentation? [U] / [CO4]**
   Internal fragmentation occurs when a fixed-sized memory block is allocated to a process, but the process uses less memory than the allocated block, leaving the unused space wasted within that block.

(**2X 5 =10 marks)**

Part B

Short Answer Questions. Answer any 5 question.s
Each question carries 6 marks

6. **Differentiate Microkernels and Module based operating system structures. [U] / [CO1]**
   Micro Kernel, contains a minimum number of functions that are required, data, and features simply to implement OS. Features of Microkernel include Minimalist design. High flexibility, Reliable and Portable - (3 marks)
   Modular Kernel, is a type of kernel in which some parts of the system core will get allocated in independent files called modules. This can be added to the system at run time. It usually requires a small amount of time to load modules. Its features include:
   Monolithic design, Low flexibility, Efficient: and Performance-oriented. -- (3 marks)
7. **Explain the functions of communications system call. [U] / [CO1]**

The communication call type facilitates: Sending or receiving messages between processes, Synchronizing actions between user processes, Establishing shared memory regions for inter-process communication and Networking via sockets.

- create, delete communication connection
- send, receive messages
- transfer status information
- Attach or detach remote devices.

Explanation of communication System Call-2 Marks, functions (-4 marks)

8. **Explain the different states of a process with a diagram. [U] / [CO2**

Explanation state Process states: New, Running State, Blocked (or waiting) State, Ready State, And Terminated state (4 marks)

Process state Diagram (2 marks)

9. **Define Round Robin (RR) scheduling and explain how it works with time quantum. [K] / [CO2]**

The round-robin (RR) scheduling algorithm is designed especially for timesharing systems. The ready queue is treated as a circular queue. The CPU scheduler goes around the ready queue, allocating the CPU to each process for a time interval of up to 1 time quantum.

Explanation of Round robin scheduling with an example.

Definition -3 marks. Explanation with example 3 marks.

10. **Describe the three types of buffering in interprocess communication [U] / [CO3]**

Three buffering in interprocess communication:

i. Zero Buffering
ii. Bounded Buffering
iii. Unbounded Buffering

Explanation 2 marks each

11. **What are Monitors? Explain in detail with its appropriate syntax [U] / [CO3]**

A monitor in operating systems is a high-level synchronization construct that helps manage mutual exclusion and coordination between concurrent processes or threads.

- A monitor ensures that only one process at a time can be active within the monitor
- The monitor‘s local variables can be accessed by local monitor functions
- Implicitly the monitor defines a mutex lock.

Syntax of a monitor:

```
monitor monitor name

{
/* shared variable declarations */
function P1 ( . . . ) { . . .
}
function P2 ( . . . ) { . . .
}
.
.
.
function Pn ( . . . ) { . . .
}
initialization code ( . . . ) { . . .
}
```

Explanation- 3 marks Syntax- 3 marks

12. **What are the different types of frame allocation strategies? Explain briefly. [U] / [CO4]**

In operating systems, frame allocation strategies determine how physical memory frames are distributed among processes. These strategies are important for efficient memory management and preventing issues like thrashing.

The main types of frame allocation strategies:

1. Equal Allocation 2) Proportional Allocation 3) Priority Allocation 4). Global Allocation 5) Local Allocation. Explanation of any three strategies give 6 marks.

(6X5=30 Marks)

Part C

Essay Questions. Answer any 2 questions
Each question carries 15 marks

13. Explain process scheduling and scheduling criteria in detail and describe different types of schedulers. [U] / [CO2]

Process scheduling is the mechanism by which an operating system selects which process in the ready queue should be executed by the CPU.

Scheduling criteria:

- CPU utilization.
- Throughput.
- Turnaround time.
- Waiting time.
- Response time.

Operating systems use three types of schedulers:

1) Long-Term Scheduler (Job Scheduler)
2) Short-Term Scheduler (CPU Scheduler)
3) Medium-Term Scheduler

Process scheduling definition -2 marks
Explanation of scheduling critriea-5 marks
Types of schedulers -8 marks

14. Discuss in detail how the Banker's Algorithm helps in avoiding deadlock [U] / [CO3]

Banker's Algorithm is a deadlock avoidance algorithm. It ensures that a system never enters an unsafe state, which could lead to deadlock. It helps avoid deadlocks by carefully examining resource requests before granting them.

Explanation Banker’s algorithm a with an example
Banker’s algorithm explanation – 10 marks
Example: 5marks

15. Compare and contrast paging and segmentation in detail. [An] / [CO4]

Paging is a memory management technique in which process address space is broken into blocks of the same size called pages (size is power of 2, between 512 bytes and 8192 bytes). Similarly, main memory is divided into small fixed-sized blocks of (physical) memory called frames and the size of a frame is kept the same as that of a page to have optimum.

Segmentation is a memory-management scheme that supports this programmer view of Memory.

Explanation paging and Paging Hardware with TLB and diagram -7 marks
Explanation of segmentation with Segmentation Hardware and diagram – 6 marks.
Comparison of paging and segmentation – 2marks

(15X2 = 30 Marks)